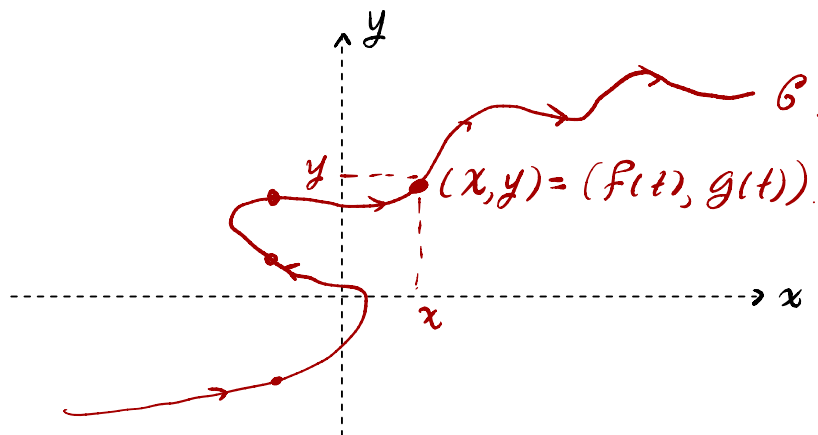


Lecture 17, Mar 06 2026

Imagine that a particle moving along the curve C .



In general C is not a function $y=f(x)$. But
Parametric equations.

Suppose that x and y are both given as functions of the variable t , called a parametric equations by the equation

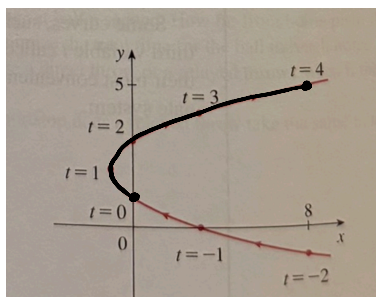
$$x = f(t) \quad y = g(t).$$

Each value t we have $(x, y) = (f(t), g(t))$ the curve is called a parametric curve.

Example. Sketch and identify the curve defined by.

$$x = t^2 - 2t, \quad y = t + 1.$$

<u>Sln.</u>	$t = -2$	$t = -1$	$t = 0$	$t = 1$	$t = 2$
	$x = 8$	$x = 3$	$x = 0$	$x = -1$	$x = 0$
	$y = -1$	$y = 0$	$y = 1$	$y = 2$	$y = 3$



Now if $0 \leq t \leq 4$. then if $t=0$ $(0,1)$ - initial point.
 if $t=4$ $(8,5)$ - terminal point.

Rmk. In general if

$$x=f(t) \quad y=g(t) \quad a \leq t \leq b.$$

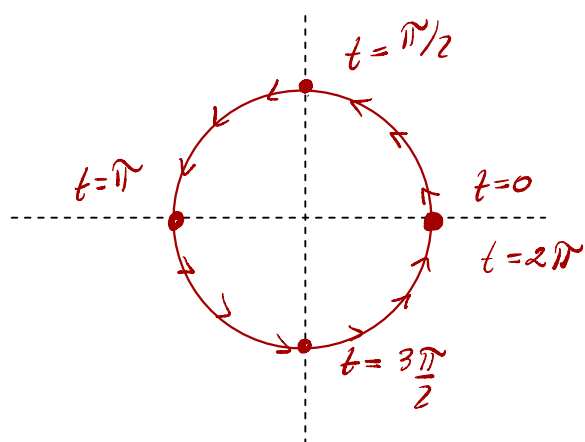
then $(f(a), g(a))$ is the initial point

$(f(b), g(b))$ is the terminal point.

Example. What curve is represented by the equations?

$$x = \cos t \quad y = \sin t \quad 0 \leq t \leq 2\pi.$$

Sln: $t=0 \Rightarrow (1,0)$; $t=\frac{\pi}{2} \Rightarrow (0,1)$; $t=\pi \Rightarrow (-1,0)$; $t=\frac{3\pi}{2} \Rightarrow (0,-1)$; $t=2\pi \Rightarrow (1,0)$



By eliminating the parameter t .

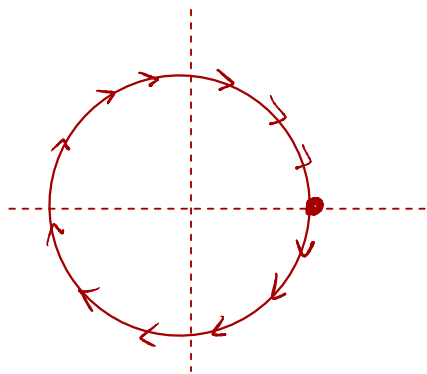
$$x^2 + y^2 = \cos^2 t + \sin^2 t = 1.$$

then

$$x^2 + y^2 = 1.$$

For example: $x = \sin 2t$, $y = \cos 2t$ $0 \leq t \leq 2\pi$.

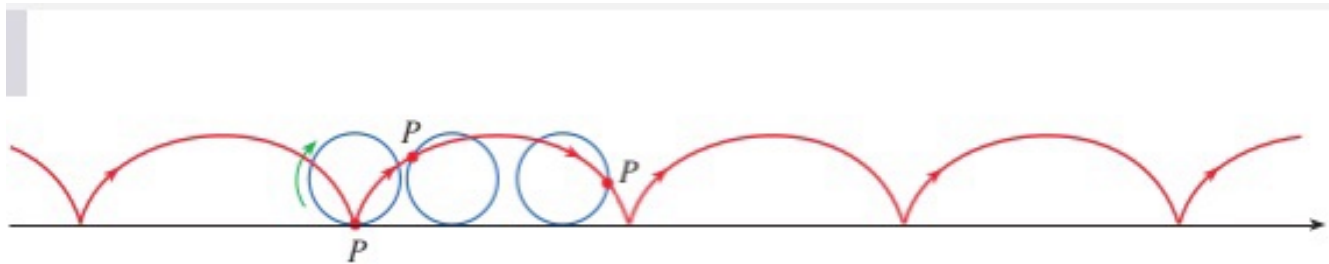
$$x^2 + y^2 = \sin^2 2t + \cos^2 2t = 1.$$



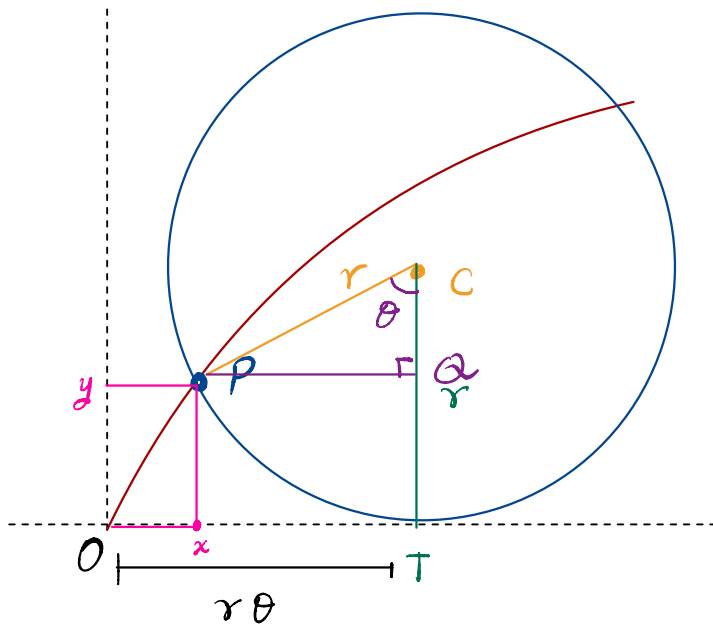
the particle moves twice around the circle
 in the clockwise direction.

The Cycloid.

Example The curve traced out by a point P on a circumference of a circle as the circle rolls along a straight line is called a cycloid. If the circle has radius r and rolls along the x -axis and if one position of P is the origin, find the parametric equations.



Sln.



Notice that

$$|OT| = \text{arc } PT = \widehat{PT} = \theta \cdot r.$$

$$x = |OT| - |PQ|$$

$$y = |TC| - |CQ|.$$

$$|OT| = r\theta$$

$$|PQ| = r \sin \theta. \quad \text{b/c.}$$

$$\sin \theta = \frac{|PQ|}{r}$$

$$|TC| = r.$$

$$|CQ| = r \cos \theta. \quad \text{b/c}$$

$$\cos \theta = \frac{|CQ|}{r}.$$

$$x = r\theta - r \sin \theta = r(\theta - \sin \theta)$$

$$y = r - r \cos \theta = r(1 - \cos \theta) \quad \theta \in \mathbb{R}.$$

10.2 Calculus with parametric equations.

* Tangents.

Let f, g be differentiable functions and

$$x = f(t), \quad y = g(t).$$

the parametric equations.

$$\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$$

then

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{\frac{d}{dt} \left(\frac{dy}{dx} \right)}{\frac{dx}{dt}}$$

Example. a). Find the tangent to the cycloid $x = r(\theta - \sin \theta)$

$$y = r(1 - \cos \theta) \quad \text{at } \theta = \pi/3.$$

b). At what points is the tangent horizontal? When is it vertical?

Sln. The slope of the tangent is

$$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta}$$

$$\frac{dy}{d\theta} = \frac{d}{d\theta} (r(1 - \cos \theta)) = \frac{d}{d\theta} (r - r \cos \theta) = 0 - r \frac{d}{d\theta} (\cos \theta) = r \sin \theta.$$

$$\frac{dx}{d\theta} = \frac{d}{d\theta} (r(\theta - \sin \theta)) = r(1 - \cos \theta)$$

$$\frac{dy}{dx} = \frac{r \sin \theta}{r(1 - \cos \theta)} = \frac{\sin \theta}{1 - \cos \theta}$$

$$\theta = \pi/3$$

$$\frac{dy}{dx} = \frac{\sin(\pi/3)}{1 - \cos \pi/3} = \frac{\sqrt{3}/2}{1 - 1/2} = \sqrt{3}$$

$$m = \sqrt{3}$$

$$x = r\left(\frac{\pi}{3} - \sin \frac{\pi}{3}\right) = r\left(\frac{\pi}{3} - \frac{\sqrt{3}}{2}\right)$$

$$y = r(1 - \cos \frac{\pi}{3}) = \frac{r}{2}$$

$$P\left(\underbrace{r\left(\frac{\pi}{3} - \frac{\sqrt{3}}{2}\right)}_{x_1}, \underbrace{\frac{r}{2}}_{y_1}\right)$$

$$y - y_1 = m(x - x_1)$$

$$y - \frac{r}{2} = \sqrt{3}\left(x - r\frac{\pi}{3} + r\frac{\sqrt{3}}{2}\right)$$

$$y = \sqrt{3}x - r\pi\frac{\sqrt{3}}{3} + \frac{3}{2}r + \frac{r}{2}$$

$$y = \sqrt{3}x + r\left(2 - \pi\frac{\sqrt{3}}{3}\right)$$